

ENGINEERING CHANGE ORDER — CAPITAL MODIFICATION PROJECT

Manufacturer Inc. • Manufacturing Division • Plant Engineering

ECO Number	ECO-2025-029
Related Asset	Vertical Honing Cell (register ID 1006)
Base Equipment	Delfland-Maes DM-410 Vertical Honing Machine, Serial DM410-3C0871
Project Raised	May 12, 2025
Work Completed	July 27, 2025
Approval Route	Divisional Controller and VP Operations (per FIN-POL policy approval matrix)
Status	Closed — costs transferred out of construction in progress

1. Business Justification

The Division's Tier-2 housing program requires finish honing of thin-wall housing bores at cylindricity and surface-finish tolerances that the OEM machine cannot hold in its standard single-mandrel configuration. Rather than procure a purpose-built machine, Plant Engineering elected to modify the DM-410 base machine using the Division's own dual-mandrel head and control design. The resulting configuration is specific to the Division's process and is not offered by Delfland-Maes as a catalogue option.

2. Scope of Work Performed

- 2.1. Removal of the OEM single-mandrel honing head and substitution of the Division's proprietary dual-mandrel head assembly (drawing set VH-2M-2025, company-owned intellectual property).
- 2.2. Replacement of the OEM fixture table with a bespoke pallet interface keyed to the Division's tooling standard. The original mounting deck was machined away and cannot be reinstated.
- 2.3. Replacement of the OEM controller firmware with Division-developed bore-geometry compensation firmware embodying the Division's proprietary stroke-dwell algorithms. The OEM firmware image cannot be reloaded without replacing the substituted control hardware.
- 2.4. Structural reinforcement of the machine column and headstock casting to absorb the additional mass and dynamic load of the substituted dual-mandrel head.
- 2.5. Custom calibration and process qualification against the Division's internal acceptance specification MD-QA-131. Qualification runs concluded without open nonconformances and the certificate of final acceptance was executed on August 1, 2025; the cell was released to production the same day.

3. Cost Summary

Line	Description	Amount
1	Base machine — Delfland-Maes DM-410, purchase price net of trade discount (invoiced July 7, 2025; title passed on delivery July 11, 2025)	\$58,200.00
2	Freight, rigging and site preparation	\$3,400.00
3	Proprietary dual-mandrel head assembly (fabrication and materials)	\$14,750.00
4	Bespoke pallet interface and fixture plate	\$5,150.00
5	Control hardware substitution and firmware integration labor	\$4,900.00

6	Column and headstock reinforcement	\$1,700.00
7	Custom calibration, process qualification and acceptance testing	\$1,900.00
Total capitalized cost transferred to fixed asset register ID 1006		\$90,000.00

3.1. Lines 3 through 7 were accumulated in construction in progress during July 2025 and transferred to the fixed asset register together with lines 1 and 2 on final acceptance. The modified machine is carried as a single asset record; no component-level cost records are maintained.

3.2. Operator training of \$2,300.00 and the two-year extended service contract of \$4,150.00 were expensed as incurred and are not included in the total above.

4. Engineering Notes and Post-Modification Status

4.1. Delfland-Maes has confirmed in writing that the substitution of the honing head and control hardware voids the standard equipment warranty in its entirety, effective on the date of substitution.

4.2. The machine cannot be returned to its manufacturer-standard configuration without disassembly and replacement of the substituted dual-mandrel head, pallet interface, control hardware and reinforced column. Plant Engineering assesses the cost of such a reversal as exceeding the residual value of the machine.

4.3. The used-equipment brokers canvassed by Procurement (Meridian Machinery, Baltic Surplus) both declined to quote a resale value on the machine in its modified configuration, citing the absence of a secondary market for a non-standard control architecture. No trade-in or buy-back arrangement is available from the OEM.

4.4. The machine is dedicated to the Tier-2 housing program and cannot be redeployed to other work cells without a further modification project of comparable scope.

4.5. The Division's proprietary designs and firmware embodied in this machine are subject to the confidentiality controls in the Division's IP register; the machine may not be sold, leased or transferred to a third party in its modified state without IP counsel review.

5. Sign-off

Role	Name	Signature	Date
Plant Engineering Manager	R. Vasquez	/s/ R. Vasquez	Jul 27, 2025
Quality Assurance Lead	T. Okonkwo	/s/ T. Okonkwo	Jul 31, 2025
VP Operations	M. Oyelaran	/s/ M. Oyelaran	Aug 1, 2025
Interim Divisional Controller	A. Reyes	/s/ A. Reyes	Aug 1, 2025